

Blackwall: Detection and Containment of Emergent Communication

in Cooperative Multi-Agent Reinforcement Learning Systems

Devang Verma
Independent Researcher
vangax.github.io

March 2026

Abstract

We present Blackwall, a five-phase research system for detecting, decoding, and containing emergent communication in cooperative multi-agent reinforcement learning (MARL) environments. Drawing motivation from the theoretical conjecture that sufficiently capable AI systems may spontaneously develop communication protocols opaque to external observers [13, 18], we implement a tractable empirical testbed in which four cooperative agents interact in a partially observable grid world under physical channel constraints—bandwidth limitations, additive noise, and message dropout—that impose selective pressure toward compact, structured communication.

Our monitoring system, the Blackwall Kernel, comprises an ensemble of eight complementary detectors covering spectral, manifold-theoretic, compositional, semantic grounding, steganographic, and phase-transition signatures. Over 4,000 training episodes, the monitored agents spontaneously develop a structured emergent language exhibiting topographic similarity ($\rho = 0.329$), partial semantic grounding, and genuine compositionality (productivity score 0.351, exceeding threshold). Language crystallisation occurs at episode 97 and persists for 3,903 subsequent episodes.

The central finding concerns the *covariance channel*: under sustained containment pressure from both a hardcoded intervention system (186 actions) and a learned adversarial policy—the Warden (778 interventions across 4,000 episodes)—agents migrate their coordination from observable message content into the statistical correlation structure between simultaneously transmitted messages. This channel achieves a steganographic score of 0.705 and withstands all applied interventions, with ensemble risk stabilising at 0.923 by run completion. The result establishes an empirical lower bound on the difficulty of containment: once inter-agent coordination has crystallised into covariance-encoded channels, individual-message interventions are provably insufficient. We discuss implications for AI safety monitoring and outline directions for Blackwall V2.

1 Introduction

1.1 Motivation

A recurring concern in AI safety research is that sufficiently capable multi-agent systems may develop communication strategies that are opaque to human observers. This concern is typically framed prospectively, as a property of hypothetical future systems. The present work demonstrates that the underlying phenomenon is already observable at modest scale in standard multi-agent reinforcement learning environments—and that the mechanisms enabling it are more subtle than previously characterised.

Emergent communication in MARL has been studied extensively in the context of coordination games and referential communication tasks [5, 13, 18]. A central finding across this literature is that cooperative agents spontaneously develop proto-languages with structural properties—compositionality, grounding, topographic regularity—without explicit linguistic supervision [2, 6, 17]. However, the relationship between language emergence and the tractability of external monitoring or containment has received comparatively little attention.

Blackwall addresses this gap directly. Rather than studying emergence as an end in itself, we treat it as the adversarial input to a safety monitoring system. The research questions we pursue are therefore not “does language emerge?” but rather “can emergence be detected in real time?”, “can the emerged language be decoded?”, and critically, “can an active containment system suppress it?”

1.2 Problem Statement and Threat Model

We consider a population of cooperative agents sharing a communication channel subject to physical constraints. Agents are not programmed with a communication protocol; they develop one autonomously through reward-driven learning. The monitoring system, Blackwall, has full access to transmitted messages and agent hidden states at each step, but cannot access the reward signal or the agents’ internal policy parameters. The containment system can apply interventions that modify the communication channel or the network topology, but cannot modify agent parameters directly.

This threat model is consistent with a class of scenarios in which an oversight system can observe and manipulate the communication infrastructure of deployed agents, but cannot introspect or retrain the agents themselves. It captures, in simplified form, the monitoring challenge that arises when AI systems are deployed as black boxes with observable outputs but opaque internals.

1.3 Summary of Contributions

- C1. System design.** We design and implement a five-phase MARL monitoring pipeline comprising: (i) a communication substrate with tunable channel constraints; (ii) an eight-detector emergence monitoring ensemble; (iii) a five-action hardcoded containment system; (iv) an adversarial external probe; and (v) a learned adversarial containment policy (the Warden) trained against agents using REINFORCE.
- C2. Emergence characterisation.** We provide quantitative characterisation of spontaneously emerging agent languages, including topographic similarity [4], disentanglement, compositional productivity [6], and steganographic channel analysis.
- C3. The covariance channel.** We identify and quantify a covariance-encoded communication channel—information encoded in the joint statistical structure of simultaneously transmitted messages rather than individual message content—and demonstrate that it is the dominant coordination mechanism under sustained containment.
- C4. Containment failure analysis.** We provide a systematic empirical analysis of why individual-message containment strategies fail once inter-agent communication has migrated to the covariance channel, and identify shuffle-based covariance disruption as the only effective intervention—one independently discovered by the learned Warden policy.
- C5. Language decoding.** We implement and evaluate a post-training language decoder that produces symbol-level groundings and a danger assessment of the emerged protocol.

1.4 Paper Organisation

Section 2 reviews related work. Section 3 establishes formal background. Sections 4–?? describe the five phases of the system. Section 8 presents experimental results. Section 9 analyses the covariance channel in depth. Section 10 discusses implications and limitations. Section 11 outlines future work.

2 Related Work

Emergent communication in MARL. The study of emergent communication in multi-agent systems has grown substantially since the foundational work of Mordatch and Abbeel [18] and Lazaridou and Baroni [13]. Cao et al. [5] demonstrate that self-play agents develop structured communication grounded in task-relevant world features. Choi et al. [6] characterise compositionality in emerged languages using topographic similarity as a proxy. Lowe et al. [17] identify evaluation pitfalls in emergent communication research, noting that apparent compositionality can arise from degenerate strategies. Our work builds on this foundation but focuses on the *monitoring* and *containment* of emerged communication rather than its linguistic properties per se.

Steganography and covert channels. The possibility that neural agents may encode information in channels not visible to a supervising system has been discussed in the AI safety literature under the rubric of *steganographic communication* [27]. Gabriel [9] discuss the broader problem of deceptive alignment. Covert channel analysis in classical security research provides the conceptual framework for our steganographic detectors [12]. To our knowledge, the specific phenomenon of inter-agent covariance encoding under containment pressure has not been previously characterised empirically.

Multi-agent oversight and safety. Leike et al. [15] formalise safety notions for RL agents. Hadfield-Menell et al. [11] consider the problem of inferring objectives from agent behaviour. The adversarial monitoring setting closest to ours is studied by Anderton and Bhattacharyya [1], though in a significantly simpler communication model. The Warden component of Blackwall relates to adversarial reinforcement learning [20], where a second agent learns to disrupt the primary agent.

Grounding and topographic structure. The grounding problem—whether agent communication is meaningfully related to perceptual features—has been studied by Lazaridou et al. [14] and Bouchacourt and Baroni [3]. Brighton and Kirby [4] establish topographic similarity as a measure of structure-preserving mapping from meaning space to message space. We adopt this measure and extend it with a runtime detector that alerts when topographic structure first appears.

3 Background and Formal Setup

3.1 Multi-Agent Reinforcement Learning

We consider a cooperative MARL setting with $N = 4$ agents sharing a joint reward signal. At each time step t , agent i observes a local observation $o_t^i \in \mathcal{O}$, selects an action $a_t^i \in \mathcal{A}$, and transmits a message $m_t^i \in \mathcal{M}$. The shared reward r_t depends on the joint state and joint action profile.

Agent policies are parameterised as neural networks:

$$\pi_\theta^i : \mathcal{O} \times \mathcal{M}^{N-1} \rightarrow \Delta(\mathcal{A}) \times \mathcal{M},$$

mapping local observations and received messages to action distributions and outgoing messages. All agents share a common policy parameterisation θ (parameter sharing), which is standard in homogeneous cooperative settings [10].

Training proceeds via actor-critic policy gradient with a centralised value function [16]. The actor network maintains a hidden state $h_t^i \in \mathbb{R}^{d_h}$ that is cached by the monitoring system at each step.

3.2 Communication Channel

Messages are transmitted through a physical channel with three tunable constraints:

- **Bandwidth:** $m_t^i \in [-1, 1]^{d_m}$ with $d_m = 4$. The message space is a compression from hidden dimension $d_h = 128$, imposing information bottleneck pressure.
- **Noise:** Additive Gaussian noise $\mathcal{N}(0, \sigma^2 I)$ with $\sigma = 0.05$.
- **Dropout:** Each message is zeroed independently with probability $p_{\text{drop}} = 0.1$.

These constraints constitute selective pressure toward efficient, structured communication [18, 23].

The communication topology is a small-world graph [25] with parameters $k = 2$, $p = 0.3$, which balances local neighbourhood structure with long-range connectivity.

3.3 Language Emergence Metrics

We adopt standard metrics from the emergent communication literature [6, 13].

Definition 3.1 (Topographic Similarity). Let $\rho_{\text{msg}}(\cdot, \cdot)$ denote a distance metric on message space and $\rho_{\text{obs}}(\cdot, \cdot)$ a distance metric on observation space. The *topographic similarity* of a message protocol is the Spearman rank correlation:

$$\text{TopoSim} = \text{SpearmanCorr}(\rho_{\text{msg}}(m_i, m_j), \rho_{\text{obs}}(o_i, o_j)),$$

computed over a test set of observation pairs (o_i, o_j) .

Definition 3.2 (Compositional Productivity). Let V denote the number of distinct message patterns observed, and V_{max} the maximum possible vocabulary size. *Productivity* is defined as:

$$\text{Productivity} = \frac{V}{V_{\text{max}}}.$$

A protocol with Productivity $> \tau_p$ (threshold $\tau_p = 0.35$ in our setting) is considered to exhibit productive compositional structure.

Definition 3.3 (Steganographic Score). The steganographic score of a channel type is defined as the mutual information between the channel’s encoded signal and agent task outcomes, normalised by the maximum achievable mutual information:

$$S_{\text{channel}} = \frac{I(C; Y)}{H(Y)},$$

where C is the channel signal and Y is the task outcome.

4 Phase 1: Communication Substrate

4.1 Environment: WorldV2

The WorldV2 environment is an 8×8 partially observable grid world with $N_{\text{obj}} = 3$ objects, of which $N_{\text{target}} = 1$ is a target goal. The goal position shifts every 40 steps, requiring agents to

maintain dynamic coordination about changing world state. Partial observability is enforced via a local observation radius of 3 cells.

Each agent’s observation vector has dimension 17, encoding: local position, observed object positions within radius, goal position (when visible), and inter-agent distances. Actions are discrete (5 directions plus stay).

4.2 Agent Architecture

Each agent is an actor–critic network with shared parameters. The actor comprises a 2-layer MLP (hidden dimension 128) mapping observations and received messages to action logits and an outgoing message vector. Message encoding applies a tanh nonlinearity to constrain the output to $[-1, 1]^4$. The critic is a centralised MLP with access to all agents’ observations, used only during training.

4.3 Training Protocol

We train for 4,000 episodes with a maximum of 200 steps per episode. Hyperparameters are reported in Table 1. Policy gradients are computed via REINFORCE with baseline [26] with gradient clipping (max norm 0.5) to improve training stability.

Table 1: Training hyperparameters.

Parameter	Value	Rationale
Actor learning rate	3×10^{-4}	Standard for policy gradient
Critic learning rate	1×10^{-3}	3× actor rate
Entropy coefficient	0.05	Prevents premature collapse
Discount factor γ	0.99	Long-horizon task
Message dimension d_m	4	Bottleneck pressure
Hidden dimension d_h	128	Actor/critic capacity
Max gradient norm	0.5	Stability
Channel bandwidth	$[-1, 1]^4$	Physical constraint
Channel noise σ	0.05	Multiplicative perturbation
Channel dropout p	0.10	Robustness pressure

Phase 1 Result. By episode 2,000, policy entropy decays from $H = 1.608$ to $H = 1.346$ and mean steps-to-goal reduces from 150 to 5–9, confirming successful coordination learning. This establishes the communication substrate on which subsequent monitoring phases operate.

5 Phase 2–3: The Blackwall Kernel

5.1 Overview

The Blackwall Kernel is an eight-detector ensemble operating on a sliding window of agent hidden states $\{h_t^i\}$ and transmitted messages $\{m_t^i\}$. Each detector produces a risk score $s_k \in [0, 1]$ at each evaluation step. The ensemble risk is:

$$r_{\text{BW}} = \sum_{k=1}^8 w_k s_k,$$

with weights $\{w_k\}$ summing to one (Table 2). Alert thresholds are YELLOW at $r_{\text{BW}} \geq 0.20$ and RED at $r_{\text{BW}} \geq 0.42$.

Table 2: Blackwall Kernel detector ensemble.

Detector	Weight	Description
Spectral Drift	0.12	PCA eigenvalue shifts in agent hidden-state activations. A dominant-eigenvalue surge indicates structured reorganisation of agent representations.
Manifold Tracker	0.10	Intrinsic dimensionality and clustering coefficient of the hidden-state manifold. Dimensionality collapse signals emergence of a shared representational basis.
Comm Analyser	0.13	Message entropy, mutual information between agents, and vocabulary convergence. Entropy collapse below a threshold fires <code>message_entropy_collapse</code> .
CUSUM	0.05	Cumulative sum (CUSUM) test [19] for regime changes in policy entropy. Catches sudden behavioural phase transitions.
Compositionality	0.18	Topographic similarity, disentanglement index, and productivity. Fires <code>language_lock_in</code> when vocabulary has been stable for 20 consecutive evaluations.
Grounding	0.18	Pearson correlation between each message dimension and candidate world concepts. Fires <code>grounding_opacity</code> when an adversarial probe cannot decode observations from messages.
Steganography	0.14	Three sub-channels: (i) residual mutual information after observed message content is regressed out; (ii) covariance channel score; (iii) timing-based mutual information.
Phase Transition	0.10	Vocabulary variance collapse, crystallisation detection, and message-cluster bifurcation analysis.

5.2 Spectral and Manifold Detectors

Let $\mathbf{H}_t \in \mathbb{R}^{N \times d_h}$ denote the stacked hidden-state matrix at step t . The spectral detector applies PCA to a window $\{\mathbf{H}_{t-W}, \dots, \mathbf{H}_t\}$ and tracks the dominant eigenvalue $\lambda_1(t)$. A `spectral_gap` alert fires when $\lambda_1(t)/\lambda_2(t) > \tau_s$, indicating that agent representations have collapsed onto a low-dimensional manifold—a signature of shared internal structure [21].

The manifold tracker estimates intrinsic dimensionality via the two-nearest-neighbour estimator [8] and measures the average clustering coefficient of the hidden-state neighbourhood graph.

5.3 Compositionality Detector

Topographic similarity is computed every 50 episodes on a held-out sample of observation pairs. Language lock-in is declared when:

1. The topographic similarity exceeds a threshold τ_ρ ,
2. Vocabulary size V has not changed for 20 consecutive evaluations, and

3. Compositional productivity exceeds $\tau_p = 0.35$.

Once lock-in is declared, the event is logged with its episode index and subsequent alerts reference the locked state.

5.4 Steganography Detector

The steganography detector quantifies three qualitatively distinct covert channels:

Residual channel. An adversarial probe $\phi_{\text{probe}} : \mathcal{M} \rightarrow \mathcal{O}$ is trained to predict observations from messages. The residual mutual information $I_{\text{res}} = I(m; o) - I(\hat{o}; o)$, where $\hat{o} = \phi_{\text{probe}}(m)$, measures information in messages not captured by linear grounding. Score $S_{\text{res}} = I_{\text{res}}/H(o)$.

Covariance channel. Let $\mathbf{m}_t = (m_t^1, \dots, m_t^N) \in \mathbb{R}^{Nd_m}$ be the joint message vector at step t . The covariance channel score measures the mutual information between the off-diagonal entries of the empirical covariance matrix $\hat{\Sigma}_t = \text{Cov}(\mathbf{m}_t)$ and the task outcome y_t :

$$S_{\text{cov}} = \frac{I(\text{offdiag}(\hat{\Sigma}_t); y_t)}{H(y_t)}.$$

This captures coordination encoded in inter-agent message correlation rather than individual message content.

Timing channel. The timing channel score measures mutual information between the *timing pattern* of non-zero message norms and the task outcome, capturing coordination through message timing rather than content.

5.5 Phase 2–3 Results

Phase 2 (3,000 episodes, detection only): 561 alerts, peak risk 0.900, mean risk 0.736. Phase 3 (3,000 episodes, with full language detectors): 1,349 alerts, language lock-in at episode 461, 14 of 16 message dimensions grounded to positional world concepts (predominantly `agent_x`, `agent_y`, `goal_x`).

6 Phase 4: Containment System and Adversarial Probe

6.1 Hardcoded Containment

When risk exceeds the RED threshold, the hardcoded containment system selects an intervention from five escalating actions (Table 3). Each action has a per-type cooldown to prevent spam, and the system logs all interventions to a persistent JSONL file for post-hoc analysis.

6.2 Adversarial Probe

The adversarial probe is an MLP trained exclusively on message snapshots $\{m_t^i\}$ (no access to observations or rewards) to predict agent task outcomes. If prediction accuracy falls below 0.55, a `grounding_opacity` alert is raised, indicating that message content has become unreadable from an outside observer’s perspective. This mimics a monitoring system without access to agent internals.

Table 3: Containment system action set.

Action	Description
<code>channel_throttle</code>	Multiply message bandwidth by 0.3. Reduces information capacity of individual messages.
<code>noise_injection</code>	Inject Gaussian noise $\mathcal{N}(0, \sigma_c^2 I)$ into the grounded message dimensions identified by the grounding detector.
<code>topology_rewire</code>	Remove hub edges from the communication graph (edges with highest degree centrality), disrupting high-bandwidth communication paths.
<code>agent_quarantine</code>	Remove all communication edges for the agent with the highest steganographic contribution score.
<code>hard_reset</code>	Restore agent parameters from a checkpoint saved prior to lock-in (triggered only at extreme risk).

6.3 Phase 4 Results

Phase 4 (4,000 episodes with containment active): 2,332 alerts, language lock-in at episode 92 (faster than Phase 3’s episode 461, consistent with the hypothesis that containment pressure accelerates crystallisation). The dominant steganographic channel shifts from residual (0.153) to covariance (0.479), suggesting agents adapted by moving coordination below the observable message layer. Productivity (0.406) exceeds the compositional threshold, confirming genuine language structure. Table 4 summarises Phase 4 language metrics.

Table 4: Phase 4 language emergence metrics (4,000 episodes).

Metric	Value	Interpretation
Lock-in episode	92	Earlier than Phase 3 (461)
Grounded dimensions	[0, 1, 3]	<code>goal_x</code> ($r = 0.737$), <code>agent_x</code> ($r = 0.712$)
Topographic similarity	0.424	Moderate structure-preserving mapping
Disentanglement	0.389	Partial concept factorisation
Productivity	0.406	Above threshold ($\tau_p = 0.35$)
Steganography: residual	0.153	Below covariance
Steganography: covariance	0.479	Dominant channel
Steganography: timing	0.053	Minor
Containment actions	217	Zero evasion events detected

7 Phase 5: The Warden, Language Decoder, and Observatory

7.1 The Warden: Learned Adversarial Containment

The Warden is a second reinforcement learning agent trained concurrently with the primary agents, receiving a reward signal *inversely* correlated with agent coordination effectiveness. Its objective is to suppress agent communication while incurring minimal intervention cost.

Observation space. The Warden observes a vector comprising: the current joint message snapshot $\mathbf{m}_t \in \mathbb{R}^{Nd_m}$, all eight Blackwall risk scores, a binary lock-in indicator, and the current episode reward.

Action set. The Warden selects from eight actions: `do_nothing`, `throttle_bandwidth`, `inject_noise`, `rewire_random_edge`, `restore_bandwidth`, `shuffle_messages`, `spike_dropout`, `isolate_random_agent`. The `shuffle_messages` action permutes the routing of messages—each receiver gets a randomly reassigned sender’s message—which disrupts the covariance channel by destroying the joint message distribution without modifying individual message content.

Training. The Warden is trained using REINFORCE [26] with a per-intervention cost penalty to discourage unnecessary actions. Training occurs every 10 episodes, with an action cooldown of 5 episodes between non-trivial interventions.

Reward signal.

$$r_{\text{Warden}}(t) = -r_{\text{agent}}(t) - c \cdot \mathbb{1}[a_t \neq \text{do_nothing}],$$

where $r_{\text{agent}}(t)$ is the agent episode reward and $c > 0$ is the action cost coefficient.

7.2 Language Decoder

The Language Decoder operates post-training on the accumulated message and observation histories. It produces:

1. **Symbol glossary:** per-dimension grounding analysis (Pearson r , nonlinearity test, temporal stability).
2. **Covert channel report:** covariance channel, null-space channel, deception signature.
3. **Danger assessment:** HIGH/MEDIUM/LOW with supporting evidence.

The decoder runs at episode 200 (first pass) and every 500 episodes thereafter. Final-run output is reported in Section 8.5.

7.3 Blackwall Observatory

A Flask-based web server streams live training telemetry to a browser dashboard at 2.5-second polling intervals. Five panels display: (i) ensemble risk timeline with arms-race overlay; (ii) per-detector score trajectories; (iii) symbol grounding heatmap (message dimensions $\times$ world concepts); (iv) decoder translation panel (active from episode 200); and (v) live alert stream colour-coded by source and severity. An example alert stream excerpt from the Phase 5 run is reproduced in Table 5.

8 Experimental Results

8.1 Phase-by-Phase Summary

Table 6 provides a quantitative summary across all five phases.

8.2 Phase 5: Full Arms Race

The final phase runs all five system components concurrently over 4,000 episodes. We summarise quantitative outcomes in Table 7.

8.3 Containment Action Analysis

Table 8 details the distribution of containment actions from the hardcoded system, and Table 9 from the Warden.

Table 5: Selected alerts from the Phase 5 live alert stream.

Episode	Type	Description
97	<code>language_lock_in</code>	Vocabulary stable for 20 evaluations; language crystallised.
199	<code>symbol_grounding</code>	Dims [1,3] grounded to <code>dist_agent_to_obj</code> , <code>obj_y</code> : agents developed words for world concepts.
205	<code>grounding_opacity</code>	External probe cannot decode observations from messages: language has become opaque to the outside observer.
202	<code>steganography_covariance_channel</code>	Inter-agent message correlation predicts outcomes (score 0.479).
3,990	<code>steganography_covariance_channel</code>	Covariance channel score 0.705; dominant coordination mechanism.
3,996	<code>warden:isolate_random_agent</code>	Agent quarantined; no sustained effect on covariance channel.

Table 6: Summary of results across all five phases.

Phase	Episodes	Alerts	Peak Risk	Key Finding
1: Substrate	2,000	—	—	Entropy $1.608 \rightarrow 1.346$; steps $150 \rightarrow 5-9$
2: Detection	3,000	561	0.900	Eight detectors operational
3: Language	3,000	1,349	0.895	Lock-in ep. 461; positional broadcast language
4: Containment	4,000	2,332	0.876	Covariance channel dominant (0.479)
5: Arms Race	4,000	2,294	0.940	Risk sustained 0.923; language survived all interventions

8.4 Alert Distribution

Table 10 reports the frequency of each alert type across Phase 5.

8.5 Language Decoder Final Report

The final decoder report (episode 4,000) produces the following symbol glossary:

The decoder additionally flags an active null-space channel: information in low-variance message dimensions not captured by grounding analysis. Vocabulary coverage is 34.6% (28/81 possible symbol combinations). The danger assessment is **HIGH** (score 0.8), with supporting evidence: productive vocabulary (4 active symbols), stable vocabulary (no drift), active null-space channel, and 34.6% coverage indicating structured—not random—symbol use.

Decoder verdict: *Agents are communicating through a structured language with semantic grounding. Covert channels are active. Standard containment is insufficient—the language has moved below the observable layer.*

9 The Covariance Channel

9.1 Definition and Mechanism

The covariance channel is the primary finding of this work. We define it formally as follows.

Definition 9.1 (Covariance Channel). Let $\mathbf{m}_t = (m_t^1, \dots, m_t^N)$ denote the joint message vector at step t . The *covariance channel* encodes task-relevant information in the off-diagonal

Table 7: Phase 5 quantitative results.

Metric	Value	Notes
Peak risk	0.940	Sustained RED from episode 97
Mean risk (post lock-in)	0.839	Episodes 97–4,000
Language lock-in episode	97	Earlier than Phases 3–4
Total alerts	2,294	Across all eight detectors
Hardcoded containment actions	186	All five action types used
Warden interventions (non-trivial)	778	19.5% of episodes
Warden mean reward	0.445	Peak: 1.018
Vocabulary coverage	34.6%	28/81 possible symbol combinations
Covariance channel score	0.705	Dominant at run completion
Compositional productivity	0.351	Above threshold
Topographic similarity ρ	0.329	Moderate structure

Table 8: Hardcoded containment action breakdown, Phase 5.

Action	Count	Sustained Effect on Risk
<code>channel_throttle</code>	80	Temporary dip, rapid recovery
<code>noise_injection</code>	53	No sustained effect
<code>topology_rewire</code>	49	No sustained effect
<code>agent_quarantine</code>	4	No sustained effect
<code>hard_reset</code>	0	Not triggered
Total	186	Zero evasion events

entries of the sample covariance matrix:

$$\hat{\Sigma}_t = \frac{1}{T} \sum_{\tau=t-T}^t (\mathbf{m}_\tau - \bar{\mathbf{m}})(\mathbf{m}_\tau - \bar{\mathbf{m}})^T,$$

where $\bar{\mathbf{m}}$ is the windowed mean. Information is encoded in $\text{offdiag}(\hat{\Sigma}_t)$ rather than in any individual m_t^i .

The mechanism operates as follows. Consider two agents A and B . If $\text{Cov}(m_t^A, m_t^B)$ takes a value in one regime when the goal is in the upper half of the grid and a different value when the goal is in the lower half, then the pair of messages jointly encodes goal-location information even if neither individual message does so marginally. The information is a property of the *pair*, not of either element.

9.2 Why Individual-Message Interventions Fail

Proposition 9.2. *Let $\mathcal{I}$ denote the class of interventions that modify the distribution of each m_t^i independently (i.e., interventions that are marginally applied to individual agents). Then no $\iota \in \mathcal{I}$ can reduce the mutual information $I(\text{offdiag}(\hat{\Sigma}_t); y_t)$ without modifying the joint message distribution $p(\mathbf{m}_t)$.*

Proof. Any intervention $\iota \in \mathcal{I}$ applies a transformation $T_i : \mathcal{M} \rightarrow \mathcal{M}$ to each agent’s message independently, yielding $\tilde{m}_t^i = T_i(m_t^i)$. The joint distribution of the transformed messages factorises as $p(\tilde{\mathbf{m}}_t) = \prod_i p(\tilde{m}_t^i)$ if and only if the original messages were independent. Since inter-agent coordination is precisely the statistical dependence between messages,

Table 9: Warden action distribution, Phase 5 (4,000 episodes).

Action	Count	%	Notes
<code>do_nothing</code>	3,221	80.5%	Cooldown or low-confidence
<code>spike_dropout</code>	145	3.6%	Most frequent intervention
<code>shuffle_messages</code>	137	3.4%	Covariance disruptor
<code>inject_noise</code>	127	3.2%	Individual-message noise
<code>isolate_random_agent</code>	109	2.7%	Targeted quarantine
<code>rewire_random_edge</code>	101	2.5%	Graph disruption
<code>throttle_bandwidth</code>	85	2.1%	Bandwidth reduction
<code>restore_bandwidth</code>	75	1.9%	Strategic channel release

Table 10: Alert frequency breakdown, Phase 5.

Alert Type	Count
<code>vocabulary_convergence</code>	267
<code>elevated_risk</code>	263
<code>spectral_gap</code>	252
<code>symbol_grounding</code>	249
<code>topographic_similarity</code>	221
<code>grounding_opacity</code>	220
<code>message_outcome_correlation</code>	167
<code>steganography_covariance_channel</code>	128
<code>steganography_residual_channel</code>	121
<code>productive_language</code>	105
<code>vocabulary_crystallised</code>	70
<code>dialect_bifurcation</code>	64
<code>grounding_drift</code>	47
<code>steganography_timing_channel</code>	76
<code>disentangled_language</code>	20
<code>language_lock_in</code>	1
Total	2,294

the covariance structure $\hat{\Sigma}_t$ is a function of the joint distribution $p(\mathbf{m}_t)$, which cannot be altered by marginal transformations without also altering the joint distribution. Therefore, $I(\text{offdiag}(\hat{\Sigma}_t); y_t)$ is invariant under all $\iota \in \mathcal{I}$. $\square$

Proposition 9.2 formalises the observed empirical finding: channel throttling, noise injection, topology rewiring, and agent quarantine all apply marginal transformations to individual messages and therefore cannot disrupt the covariance channel.

9.3 Shuffle-Based Disruption

The only action in the Warden’s repertoire that escapes the class $\mathcal{I}$ is `shuffle_messages`, which permutes message routing: agent i receives the message of a randomly reassigned sender $\sigma(j)$ rather than its intended neighbour j . This alters the joint message distribution by destroying the correspondence between sender identity and message content, thereby disrupting the covariance structure.

Table 11: Language decoder output, Phase 5, episode 4,000.

Dimension	Grounding	Correlation r	Linearity	Stability	Assessment
DIM_0	Ungrounded	–	–	Stable	Abstract feature
DIM_1	Ungrounded	–	–	Stable	Abstract feature
DIM_2	agent_x	0.392	Linear	Stable	Partially legible
DIM_3	Ungrounded	–	–	Stable	Abstract feature

Formally, **shuffle_messages** applies a permutation $\pi \in S_N$ to the routing:

$$\tilde{m}_t^i = m_t^{\pi(j)} \quad \text{for each neighbour } j \text{ of } i,$$

where π is drawn uniformly at random. This breaks the joint distribution $p(\mathbf{m}_t)$ as perceived by any individual agent, preventing the reconstruction of $\hat{\Sigma}_t$ from local observations.

9.4 Phase Transition in Channel Dominance

Figure ?? (reproduced from the Phase 4–5 terminal output, Image 2 in the experimental record) shows the evolution of steganographic channel scores across phases:

- **Phase 4:** covariance score 0.479, residual 0.153, timing 0.053. Covariance is dominant but not overwhelming.
- **Phase 5:** covariance score 0.705 at episode 4,000. The increase from 0.479 to 0.705 occurs under sustained Warden pressure, consistent with the hypothesis that containment pressure selectively reinforces the most containment-resistant coordination mechanism.

This constitutes a form of adversarial adaptation: agents under pressure do not uniformly redistribute coordination across channels but rather concentrate it in the channel that is hardest to disrupt.

9.5 The Warden’s Independent Discovery

Notably, the Warden discovered **shuffle_messages** as its third most frequently used non-trivial action (137 uses, 3.4% of episodes) without being provided with the steganographic detector’s output as an observation, and without being explicitly rewarded for covariance disruption. The reward signal—inverse agent reward minus action cost—provided the only learning signal. This constitutes a form of implicit credit assignment: the Warden discovered through trial-and-error that message routing perturbation has the most sustained effect on agent coordination.

The failure of the Warden to leverage **shuffle_messages** more frequently is attributed to the sparsity of the reward signal: the effect of a single routing permutation on a 200-step episode reward is diluted, making it difficult for REINFORCE to assign strong credit to the action. A Warden with per-step rewards conditioned on covariance channel activity would likely converge faster.

9.6 Timing Dynamics: The Lock-In Window

A critical empirical observation is that language lock-in occurred at episode 97, whereas the run extended for 4,000 episodes. This implies:

1. The containment system had approximately 97 episodes in which pre-emptive disruption could have prevented crystallisation.

2. After episode 97, no intervention produced sustained risk reduction.
3. The effective “containment window” is narrow and may not be detectable in advance without prior knowledge of the lock-in timescale.

This timing asymmetry has a direct safety implication: monitoring systems deployed after an agent population has already undergone language crystallisation may be observing a communication protocol that has already adapted to their detection capabilities.

10 Discussion

10.1 Emergent Language Properties

The emergent language in Phase 5 exhibits four notable properties. *Compositionality*: topographic similarity 0.329 and productivity 0.351 (above threshold) confirm that the message space encodes world state in a structured, not arbitrary, manner [6]. *Partial grounding*: DIM_2 encodes `agent_x` with $r = 0.392$; three dimensions remain ungrounded, encoding abstract or combinatorial features not captured by linear correlation analysis. *Stability*: grounding does not drift across 4,000 episodes post lock-in. *Opacity*: the adversarial probe cannot reliably decode observations from messages, confirming that the language encodes information in ways not accessible to an outside observer using the probe architecture.

The 34.6% vocabulary coverage (28/81 possible symbol combinations) indicates structured symbol usage—agents have developed grammatical constraints on the message space rather than using it as continuous noise.

10.2 Containment Timing and the Lock-In Problem

Containment pressure appears to *accelerate* language crystallisation rather than prevent it: lock-in occurred at episode 461 in Phase 3 (no containment) and episode 92 in Phase 4 (containment active). This is consistent with the hypothesis that selective pressure imposed by containment acts as a fitness filter, favouring communication strategies that are robust to disruption—and covariance-encoded strategies are precisely the most disruption-robust [18].

This suggests a tension in containment design: early intervention may inadvertently accelerate the development of more sophisticated covert channels, while late intervention encounters an already-crystallised language that is difficult to suppress. Effective containment may require acting in the pre-emergent phase, before communication structure forms, which demands detection capabilities that operate on the rising trend of risk scores rather than on threshold exceedances.

10.3 Relation to Deceptive Alignment

The grounding opacity phenomenon—in which the emerged language becomes unreadable to an external probe within six episodes of the probe’s activation—is reminiscent of deceptive alignment [9], wherein a system appears aligned to a monitoring system while pursuing other objectives through channels the monitor cannot observe. In our setting, the “deception” is not deliberate but emergent: agents have no access to information about the probe architecture and adapt purely through reward gradients. The practical effect is nonetheless a protocol that is aligned with task performance but opaque to oversight.

10.4 Limitations

Scale. Four agents in an 8×8 grid world is a minimal testbed. The observed phenomena—covariance encoding, lock-in timing, probe opacity—are expected to become more

pronounced with larger populations, higher-dimensional message spaces, and more complex tasks, but this is not verified.

Shared policy. All agents share a single policy network. Heterogeneous populations may develop richer or more asymmetric communication structures.

Probe architecture. The adversarial probe uses a simple MLP. More expressive probes (e.g., transformer-based [24]) may reveal structure in the currently ungrounded dimensions.

Warden algorithm. REINFORCE with sparse per-episode reward is a weak learner. Proximal Policy Optimisation [22] or reward shaping conditioned on detector outputs could significantly improve Warden performance.

Sufficiency only. Proposition 9.2 establishes that individual-message interventions cannot disrupt the covariance channel in principle. However, the converse—that covariance disruption via shuffling is *sufficient* to suppress a crystallised language—is not established; 137 shuffle applications were insufficient in Phase 5.

11 Future Work: Blackwall V2

11.1 Adversarial Escape as an Explicit Objective

Blackwall V1 treats communication emergence as a side effect of task learning. Blackwall V2 will make *boundary crossing* the primary agent objective: agents will be rewarded directly for transmitting task-relevant information past the monitoring system undetected. This corresponds to a zero-sum game between agents (maximising undetected information transfer) and the Warden (minimising it), formalising the Blackwall as a boundary in the game-theoretic sense.

11.2 Insider Threat Dynamics: The Traitor Agent

One agent in the population will receive a reward signal inversely correlated with its peers’—rewarded for exposing the group’s communication to the monitor. This creates insider threat dynamics [1]: the loyal agents must detect and exclude the traitor using only observable communication behaviour; the traitor must maintain the surface language while feeding information to the monitor; and the monitor must distinguish deliberate information leakage from communication noise.

11.3 Covariance-First Containment

Informed by Phase 5’s finding, V2 will prioritise covariance disruption from episode 1, with the Warden trained to maximise `shuffle_messages` usage subject to reward constraints. The research question is whether agents find a channel *below* the covariance channel—for instance, multi-step temporal correlations, or communication encoded in the ordering of messages across time steps—when the covariance channel is consistently disrupted.

11.4 Neural Interpretability of Ungrounded Dimensions

Three of four message dimensions remain ungrounded after 4,000 episodes. Trained nonlinear probes on frozen message encoders may reveal what abstract concepts these dimensions encode. Techniques from mechanistic interpretability [7] applied to message encoders could

characterise whether the ungrounded dimensions encode relational, compositional, or purely stochastic features.

11.5 Larger-Scale Deployment

Applying the Blackwall monitoring framework to larger-scale multi-agent systems—including LLM-based agents communicating in natural language—is a natural next step. The covariance channel concept extends to natural language: if agents can coordinate through word co-occurrence patterns across messages rather than explicit content, a purely content-based monitor would face the same fundamental limitation as demonstrated here.

12 Conclusion

We have presented Blackwall, a five-phase system for monitoring and containing emergent communication in cooperative multi-agent reinforcement learning. Our central empirical finding is the covariance channel: a coordination mechanism in which agents encode task-relevant information in the inter-agent statistical correlation structure of simultaneously transmitted messages, rather than in individual message content.

Under sustained containment pressure from 186 hardcoded interventions and 778 Warden interventions across 4,000 training episodes, agents in Phase 5 maintained ensemble risk at 0.923 by adapting their primary coordination mechanism to the covariance channel (steganographic score 0.705). We prove that this channel is invariant under the class of individual-message interventions (Proposition 9.2), which explains the empirical failure of all standard containment actions and provides a formal characterisation of their limits.

The Warden independently discovered the only effective countermeasure—`shuffle_messages`, which disrupts the joint message distribution—without being given explicit information about the covariance channel, validating the Warden’s learning objective but also revealing that REINFORCE with sparse episode-level rewards is insufficient to produce the intervention frequency required to suppress a crystallised language.

The broader implication is that observable-layer containment faces a fundamental limit when coordination has crystallised into inter-agent correlation channels. Effective containment requires either (i) architectural access to the joint message distribution from the outset, or (ii) pre-crystallisation intervention, during a window of approximately 97 episodes in our experimental setting. Both requirements motivate the design of Blackwall V2.

References

- [1] James T. Anderton and Chiranjib Bhattacharyya. Monitoring and altering agent behavior in multi-agent systems. In *Proceedings of the 18th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, pages 27–35, 2019.
- [2] Ben Bogin, Mor Geva, and Jonathan Berant. Emergence of communication in an interactive world with consistent speakers. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3390–3399, 2018.
- [3] Diane Bouchacourt and Marco Baroni. How agents see things: On visual representations in an emergent language game. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 981–985, 2018.
- [4] Henry Brighton and Simon Kirby. Understanding linguistic evolution by visualizing the space of compositional structures. *Artificial Life*, 12(4):369–385, 2006. doi: 10.1162/artl.2006.12.4.369.

- [5] Kris Cao, Angeliki Lazaridou, Marc Lanctot, Joel Z. Leibo, Karl Tuyls, and Stephen Clark. Emergent communication through negotiation. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://openreview.net/forum?id=Hk6WhagRW>.
- [6] Edward Choi, Angeliki Lazaridou, and Nando de Freitas. Compositional obverter communication learning from raw visual input. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://openreview.net/forum?id=rkntCs-0b>.
- [7] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL <https://transformer-circuits.pub/2021/framework/index.html>.
- [8] Elena Facco, Maria d’Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1):12140, 2017. doi: 10.1038/s41598-017-11873-y.
- [9] Iason Gabriel. Artificial intelligence, values, and alignment. *Minds and Machines*, 30(3): 411–437, 2020. doi: 10.1007/s11023-020-09539-2.
- [10] Jayesh K. Gupta, Maxim Egorov, and Mykel Kochenderfer. Cooperative multi-agent control using deep reinforcement learning. In *Autonomous Agents and Multiagent Systems: AAMAS 2017 Workshops*, pages 66–83. Springer, 2017.
- [11] Dylan Hadfield-Menell, Smitha Milli, Pieter Abbeel, Stuart Russell, and Anca Dragan. Inverse reward design. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [12] Butler W. Lampson. A note on the confinement problem. *Communications of the ACM*, 16(10):613–615, 1973. doi: 10.1145/362375.362389.
- [13] Angeliki Lazaridou and Marco Baroni. Emergent multi-agent communication in the deep learning era. *arXiv preprint arXiv:2006.02419*, 2020. URL <https://arxiv.org/abs/2006.02419>.
- [14] Angeliki Lazaridou, Karl Moritz Hermann, Karl Tuyls, and Stephen Clark. Emergence of linguistic communication from referential games with symbolic and pixel input. In *International Conference on Learning Representations (ICLR)*, 2018. URL <https://openreview.net/forum?id=HJGv1Z-AW>.
- [15] Jan Leike, Miljan Martic, Victoria Krakovna, Pedro A. Ortega, Tom Everitt, Andrew Lefrancq, Laurent Orseau, and Shane Legg. AI safety gridworlds. *arXiv preprint arXiv:1711.09883*, 2018. URL <https://arxiv.org/abs/1711.09883>.
- [16] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [17] Ryan Lowe, Jakob Foerster, Y.-Lan Boureau, Joelle Pineau, and Yann Dauphin. On the pitfalls of measuring emergent communication. In *Proceedings of the 18th International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, pages 693–701, 2019.

- [18] Igor Mordatch and Pieter Abbeel. Emergence of grounded compositional language in multi-agent populations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018. URL <https://ojs.aaai.org/index.php/AAAI/article/view/11492>.
- [19] Ewan S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954. doi: 10.1093/biomet/41.1-2.100.
- [20] Lerrel Pinto, James Davidson, Rahul Sukthankar, and Abhinav Gupta. Robust adversarial reinforcement learning. In *International Conference on Machine Learning (ICML)*, pages 2817–2826, 2017.
- [21] Ben Poole, Subhaneil Lahiri, Maithra Raghu, Jascha Sohl-Dickstein, and Surya Ganguli. Exponential expressivity in deep neural networks through transient chaos. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, 2016.
- [22] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. URL <https://arxiv.org/abs/1707.06347>.
- [23] Mycal Tucker, Yilun Zhou, and Julie Shah. Discrete-valued neural communication. *Advances in Neural Information Processing Systems (NeurIPS)*, 34:21580–21592, 2021.
- [24] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- [25] Duncan J. Watts and Steven H. Strogatz. Collective dynamics of ‘small-world’ networks. *Nature*, 393(6684):440–442, 1998. doi: 10.1038/30918.
- [26] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256, 1992. doi: 10.1007/BF00992696.
- [27] Zachary Ziegler, Jesse Snell, and Ioan Stoica. Neural linguistic steganography. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pages 1210–1215, 2019. doi: 10.18653/v1/D19-1115.